DIPARTIMENTO DI INGEGNERIA E SCIENZA DELL'INFORMAZIONE

– KNOWDIVE GROUP –

KG 2025 - Territory and Health Facilities

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1 Introduction

Reusability is one of the main principles in the Knowledge Graph (KG) development process defined by iTelos. The KG project documentation plays an important to enhance the reusabiltiy of the resources handled and produced during the process. A clear description of the resources, the process (and sub processes) developed and evaluation at each step of the process provides a clear understanding of the project, thus serving such an information to external readers for the future exploitations of the project's outcomes.

The current document aims to provide a detailed report of the project developed following the iTelos methodology. The report is structured, to describe:

- Section 2: Definition of the project's purpose and related information gathering.
- Sections 3, 4, 5, 6: The description of the iTelos process phases and their activities, divided by knowledge and data layer activities, as well as the evaluation of the resources produced in terms of fit for the chosen purpose.
- Section 7: The description of the metadata produced for all (and all kind of) the resources handled and generated by the iTelos process, while executing the project.
- Section 8: Conclusion and open issues summary.

2 Project Design

The primary goal of this project is to build a **Knowledge Graph (KG)** that stores, organizes, and provides easy access to information about **essential geographical features and various healthcare facilities** within a specific territory, namely the **Trentino region**. The KG will function as a centralized knowledge hub, connecting citizens, public health planners, and emergency services to critical information regarding the **location, specialization, availability, and accessibility** of healthcare resources across the region.

The **Domain of Interest (DoI)** for the "Territory and Health Facilities" project is the **Trentino region**, focusing specifically on the relationship between its **geographical features** and **healthcare infrastructure**.

The DoI encompasses the following key aspects:

- **Geographical Scope:** The Trentino region (e.g., municipalities, towns, and locations) is the territory under study.
- **Target Entities:** Essential geographical features and different healthcare facilities (hospitals, clinics, specialized centers, emergency medical facilities) within that territory.
- **Project Focus:** Modeling the location, specialization, availability, and, crucially, the accessibility/proximity of these facilities relative to geographical points (towns, given locations).

3 Purpose Definition

3.1 Purpose Formalization

3.1.1 Scenarios Definition

In this section, a set of usage scenarios, describing the multiple aspects considered by the project purpose (linking geographical features to healthcare infrastructure), are presented.

1. **Emergency Proximity A tourist**, Marco, is hiking in a remote area (he only knows his GPS coordinates). He has a minor emergency and needs to find the absolute nearest hospital and pharmacy, regardless of the municipality.
2. **Local Specialized Care Access** A caregiver, Giulia, lives in the municipality of 'Cles'. She needs to find a semi-residential facility for her elderly father. She wants to find all options within her municipality first, and then expand her search to nearby municipalities if necessary.
3. **Service Gap & Distribution Analysis** A public health analyst, Dr. Rossi, is tasked with identifying potential "service deserts." She needs to map the locations of all psychiatric services to see which municipalities have no local access.
4. **Territorial Capacity Planning** A public health planner, Luca, is reviewing the total capacity for long-term care. He needs to know the total number of beds available in residential facilities for the elderly, aggregated by geographical area (e.g., all facilities in the municipality of 'Trento').

3.1.2 Personas

1. **Marco Bianchi** is a 32-year-old tourist visiting Trentino for a hiking trip. His main concern is safety in unfamiliar territory. He needs to be able to find the nearest hospital or pharmacy based on his phone's GPS coordinates in case of an emergency, rather than searching by a town name he doesn't know.
2. **Giulia Verdi** is a 52-year-old resident of Cles. Her main task is finding a suitable care facility for her elderly father. She needs to find all residential or semi-residential facilities that specifically offer Elderly Assistance within her home municipality.
3. **Dr. Elena Rossi** is a 45-year-old public health analyst. Her main task is creating reports on healthcare access and equity. She needs to identify which municipalities do not have a local facility offering specialized services like Psychiatric Assistance to map service deserts.
4. **Luca Verdi** is a 38-year-old health system planner. His main task is managing resource allocation for the upcoming year. He needs to know the total number of beds available in residential facilities aggregated by municipality to assess regional capacity.

3.1.3 Competency Questions (CQs)

Initally, we came up with some scratch competency questions that were not based on actual data, but an imaginary idea of what the datasets and data could include. After finalizing the datasets chosen for this project, the questions were either modified or deleted to fit the datasets better, this included an improved set of competency questions that are more precise and helpful. (make more formal, questions include emergency details that got terminated later, usage of ontological aspects)

Table 1: *Competency Questions (CQs)*

Person	No.	Question
Marco Bianchi	1.1	What is the nearest hospital to a current GPS coordinates?
Marco Bianchi	1.2	Find the address and phone number of a specific health structure.
Marco Bianchi	1.3	List all pharmacies within a 5km radius of a specific health facility.
Giulia Verdi	2.1	Which residential facilities for Elderly Assistance are located in a specific municipality?
Giulia Verdi	2.2	Are there any semi-residential structures in a specific municipality? If not, what are the closest ones?
Giulia Verdi	2.3	What specialized services are available at the health structures in a specific municipality?
Dr. Elena Rossi	3.1	Which municipalities do not have any health structure offering Psychiatric Assistance?
Dr. Elena Rossi	3.2	Generate a list of all municipalities that have more than 5 pharmacies.
Dr. Elena Rossi	3.3	What types of hospitals are present in comparison between different municipalities?
Luca Verdi	4.1	What is the total number of beds for elderly care in a municipality?
Luca Verdi	4.2	What is the total capacity of all semi-residential structures in the province?
Luca Verdi	4.3	How many health structures are managed directly by the health service in a specific municipality?

3.1.4 Concepts Identification

This section outlines the key concepts identified from the data sources and user requirements, categorizing them into Entity Types (Etypes) and defining their attributes and roles within the Knowledge Graph.

Etypes and Attributes

The identified entities are classified into three categories:

- **Core:** Central entities that directly represent the primary domain of interest (health facilities).
- **Common:** Entities that provide general context or serve as containers for other entities (geography, administrative bodies).

- **Contextual:** Entities that provide specific descriptive details or classifications for the core entities.

Table 2: *Concept Identification: Etypes and Attributes*

Entity Type	Category	Role	Attributes
HealthFacility	Core	Represents the central concept of any physical location providing healthcare services.	StructureCode (ID), Name, Address, CAP, Phone, Email, Website, Lat/Long, ServiceName
Hospital	Core	Represents a specialized facility for comprehensive medical care.	Inherits HealthFacility + ASL_Code, ManagementType
Dispensary	Core	Represents a facility for dispensing medication.	IVA (VAT), DispensaryType (Pharmacy/Parapharmacy)
CareStructure	Core	Represents a facility for residential or semi-residential care.	NumBeds, NumUsers, NumAdmissions, CareType, ManagementType
RehabStructure	Core	Specialized facility for therapeutic and rehabilitation services.	ASL_Code
Municipality	Common	Represents the administrative territory.	MunicipalityCode (ID), Name, Province

Relationships

The semantic relationships define how the identified entities interact to form the Knowledge Graph structure. These relationships are categorized into inheritance (class hierarchy) and semantic links (object properties).

Inheritance

These relationships define the specialization hierarchy, where specific facility types inherit attributes from the central `HealthFacility` class.

- **Hospital** → `subClassOf` → **HealthFacility**
- **Dispensary** → `subClassOf` → **HealthFacility**
- **CareStructure** → `subClassOf` → **HealthFacility**
- **RehabilitationStructure** → `subClassOf` → **HealthFacility**

Semantic Links (Object Properties)

These relationships connect the core entities to the common and contextual entities, establishing the domain context.

- **HealthFacility** → `locatedIn` → **Municipality**: Links a facility to its geographical territory.

3.1.5 ER Model

The ER model provides a structured foundation for implementing the KG, defining core entities and relationships based on the Competency Questions and the provided datasets. Our model is designed around a superclass entity ($HealthFacility$), which holds all common territorial data (address, coordinates, etc.), with specialized sub-entities inheriting from it to add specific attributes.

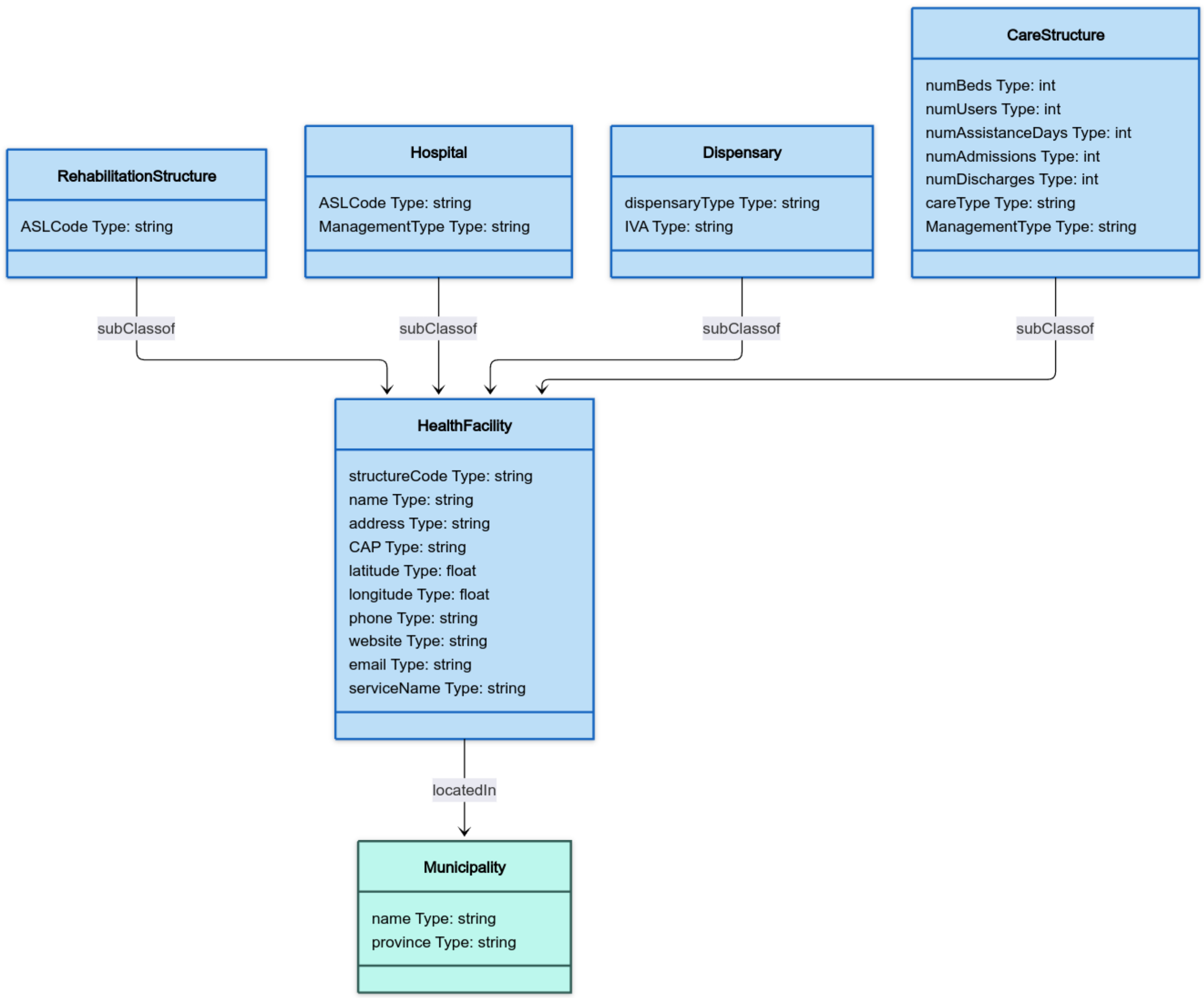

Figure 1: Entity Relation Diagram

3.2 Information Gathering

In this section, an overview of the primary input data sources available for the project is provided. This includes a list of resources that span across languages, schemas, and data values.

3.2.1 Knowledge Layer

This section outlines the semantic resources identified and utilized to structure the Knowledge Graph. For this project, Schema.org was selected as the exclusive reference ontology. This decision was driven by the need for a lightweight, highly interoperable, and web-standard vocabulary that could sufficiently model both the healthcare and territorial aspects of the domain without the excessive complexity of clinical-specific standards like FHIR.

Schema.org Vocabulary provided the foundational class hierarchy and property definitions for the majority of the entities in the Knowledge Graph. Its broad coverage allowed for a consistent modeling approach across different domains:

- **Healthcare Facilities:** The $schema : MedicalOrganization$ class served as the semantic backbone for the core $HealthFacility$ entity. Specific facility types were mapped directly to their Schema.org equivalents, such as $schema : Hospital$ for hospitals and $schema : Pharmacy$ for dispensaries.
- **Territory and Location:** The $schema : AdministrativeArea$ class was used to model municipalities, providing a standard way to represent administrative boundaries. Furthermore, the $schema : Place$ and $schema : GeoCoordinates$ (using latitude and longitude properties) were utilized to model the physical location of every facility, ensuring compatibility with standard geospatial querying tools.
- **Organization and Attributes:** Common attributes found in the source datasets, such as addresses, tax IDs, and contact information, were mapped to standard Schema.org properties like $schema : address$, $schema : vatID$, and $schema : telephone$.

3.2.2 Data Layer

Hospital Resource The Hospital Resource is a data value dataset that provides data on hospital facilities, specifically focusing on those within the Trentino region. This resource includes detailed administrative and location metrics for each facility, offering insights into the type of hospital management and contact information. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_OSP (int64):** The unique administrative code for the hospital.
- **OSPEDALE (string):** The official name of the hospital facility.
- **INDIRIZZO (string):** The street address of the facility.
- **COMUNE (string):** The municipality where the hospital is located.

- **TIPO_OSP (string):** Description of the hospital type (e.g., "OSPEDALE A GESTIONE DIRETTA").
- **TELEFONO (int64):** Contact phone number.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates of the facility.

Pharmacy Resource The Pharmacy Resource is a data value dataset that provides information on pharmacies, specifically focusing on establishments within the Trentino region. This resource identifies pharmacies by their administrative codes and provides their precise location and operational start dates. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_FARMACIA (int64):** The unique administrative code for the pharmacy.
- **FARMACIA (string):** The name of the pharmacy.
- **TIPOLOGIA (string):** The classification of the pharmacy (e.g., "ORDINARIA").
- **DATA_INIZIO (string):** The date when the pharmacy commenced operations.
- **INDIRIZZO (string):** The street address.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates.

Para-pharmacy Resource The Para-pharmacy Resource is a data value dataset that provides information on para-pharmacies (facilities authorized to sell non-prescription drugs), specifically focusing on those within the Trentino region. Similar to the Pharmacy resource, it provides location and identity data essential for mapping the pharmaceutical network. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_PARAFARMACIA (int64):** The unique code for the para-pharmacy.
- **PARAFARMACIA (string):** The commercial name of the para-pharmacy.
- **PARTITA_IVA (int64):** The VAT identification number.
- **INDIRIZZO (string):** The physical location.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates.

Health Structures Resource The Health Structures Resource is a data value dataset that provides detailed information on general health facilities, such as polyclinics, laboratories, and outpatient centers, within the Trentino region. This dataset is particularly rich in classification data, describing the type of assistance provided and the structure's operational model. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_STRUTTURA (int64):** The unique structure identifier.
- **STRUTTURA (string):** The name of the health facility.
- **TIPO_STRUTTURA (string):** The structural classification (e.g., "AMBULATORIO E LABORATORIO").
- **ASSISTENZA (string):** The type of assistance provided (e.g., "ATTIVITA‘ CLINICA", "DIAGNOSTICA").
- **SITO_WEB (string):** The website of the structure.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates.

Rehabilitation Structures Resource The Rehabilitation Structures Resource is a data value dataset that provides information about rehabilitation centers, specifically focusing on therapeutic communities and similar facilities in Trentino. While smaller in volume compared to other resources, it is critical for mapping specialized care. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **COD_RIA (int64):** The unique code for the rehabilitation structure.
- **STRUTTURA (string):** The name of the rehabilitation facility.
- **TELEFONO (int64):** Contact number for the facility.
- **LATITUDINE_P / LONGITUDINE_P (float):** The geographic coordinates.

Residential Assistance Resource The Residential Assistance Resource is a data value dataset that provides utilization and capacity data for residential facilities (e.g., RSA - nursing homes), specifically focusing on assistance to the elderly and vulnerable groups in Trentino. Unlike the registry datasets, this resource contains dynamic metrics regarding usage. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **STRUTTURA (string):** The name of the residential facility.
- **ASSISTENZA (string):** The type of assistance provided (e.g., "ASSISTENZA AGLI ANZIANI").
- **ANNO (int64):** The reference year for the data.
- **NUM_UTENTI (int64):** The total number of users served.
- **NUM_POSTI (int64):** The capacity (number of beds/places) of the structure.
- **NUM_GG_ASSISTENZA (int64):** The total days of assistance provided.

Semi-residential Assistance Resource The Semi-residential Assistance Resource is a data value dataset that provides information about day centers and semi-residential facilities. It mirrors the structure of the Residential Assistance resource but focuses on facilities where patients do not reside permanently. The original dataset was found at the Open Data Trentino portal.

Key attributes in the dataset are:

- **STRUTTURA (string):** The name of the semi-residential facility.
- **TIPO_STRUTTURA (string):** The type of structure (e.g., "STRUTTURA SEMIRESIDENZIALE").
- **NUM_UTENTI (int64):** The number of users served.
- **NUM_GIORNATE (int64):** The total number of assistance days provided.

3.3 Report

3.3.1 Personas and Scenarios

We maintained the same core scenarios and personas throughout the process, refining them to better match the specific data attributes found (e.g., focusing on specific services like "Psychiatric Assistance" because that data was available).

3.3.2 Competency Questions (CQs)

Initially, our project used a set of preliminary scratch competency questions that were based on an imagined or ideal concept of the data we might have, rather than being drawn from actual information.

Once the final datasets were selected and confirmed for the project, we conducted a crucial review to ensure these questions matched the data's real content and limitations.

Adjustment: Questions were either reworded or entirely removed to be directly relevant and answerable using the chosen data.

1.
   - **Old:** "Which municipalities in the Val di Fassa are more than a 30-minute drive from the nearest Type 1 Emergency Room?"
   - **New (Dr. Elena Rossi 3.1):** "Which municipalities do not have any health structure offering Psychiatric Assistance?"
   - **Change:** The specific focus on "Val di Fassa" and "30-minute drive" was removed because the datasets did not support real-time travel time calculations. The question was broadened to a province-wide gap analysis based on available service attributes (ASSISTENZA).
2.
   - **Old:** "What specialized services (e.g., pediatrics, cardiology) are available at the primary hospital serving the remote communities of Val di Fassa?"
   - **New (Giulia Verdi 2.3):** "What specialized services are available at the health structures in a specific municipality?"

- **Change:** The question was generalized to query any municipality, not just Val di Fassa, and simplified to match the ASSISTENZA column in the SANSTRUT001 dataset rather than inferring a "primary hospital" relationship which wasn't explicitly defined.

3. - **Old:** "Which local clinics (Poliambulatori) within 10km of Riva del Garda are currently accepting new primary care patients?"
   - **New (Marco Bianchi 1.3):** "List all pharmacies within a 5km radius of a specific health facility."
   - **Change:** The original question was dropped because the data sources provided no information on "accepting new patients" (real-time status). It was replaced with a feasible proximity query for pharmacies, which is supported by the coordinate data.

4. - **Old:** "What is the closest Trauma Center to the road intersection at Passo Pordoi, and what is the calculated ETA via helicopter transport?"
   - **New (Marco Bianchi 1.1):** "What is the nearest hospital to a current GPS coordinates?"
   - **Change:** Calculation of "ETA via helicopter" was impossible with the static facility data. The question was simplified to a direct geodesic distance calculation ("nearest hospital") using the LATITUDINE/LONGITUDINE fields.

5. - **Old:** "Which nearby hospitals are confirmed to have a functional medical evacuation helicopter pad and an available surgical team right now?"
   - **New (Luca Verdi 4.3):** "How many health structures are managed directly by the health service in a specific municipality?"
   - **Change:** Real-time resource availability ("available surgical team right now") is not in the Open Data. The question was replaced with a structural/administrative query relevant to the Planner persona (Luca Verdi).

6. - **Old:** "Which Dialysis Centers within 50km of San Martino di Castrozza offer short-term appointment slots for visiting patients?"
   - **New (Luca Verdi 4.1):** "What is the total number of beds for elderly care in a municipality?"
   - **Change:** "Appointment slots" implies a booking system, which doesn't exist in the data. The question was shifted to querying capacity (NUM_POSTI), which is a hard metric provided in the Residential Care dataset.

This focused process yielded an improved and refined set of competency questions that are now more exact, useful, and clearly supported by the information in our final data sources.

3.3.3 Entity Relationship Diagrams

During the information gathering process, the ER diagram was adjusted from a previously extremely normalized manner to a general sense fitting more onto the project itself. Main changes were integrating a "Service" entity into the "HealthFacility" entity due to its flexibility, and integrating "Management" entity into the "Hospital" entity. Also we changed the Attribute data type structure to make it more clear, each and every attribute changed from this format "(Type)(Name)" to "(Name) Type: (Type)".

4 Language Definition

The Language Definition phase represents a critical step in the iTelos methodology, serving as the linguistic and semantic bridge between the raw data gathered in the Information Gathering phase and the formal Knowledge Definition phase. Its primary objective is to resolve ambiguities, standardise terminology, and ensure that the concepts represented in the Knowledge Graph are semantically consistent and interoperable.

In the context of the "Territory and Health Facilities" project, this phase addresses the challenge of harmonizing heterogeneous datasets that primarily use Italian administrative terminology (e.g., Azienda Sanitaria, Partita IVA) with international semantic standards (e.g., HealthAuthority, vatID). By explicitly defining and fixing the language terms for Entity Types (Etypes), Object Properties, and Data Properties, we ensure that the resulting Knowledge Graph is not only accurate to the local domain but also globally comprehensible.

The activities in this phase are structured into two main layers: the Knowledge Layer, which focuses on finalising the formal vocabulary and aligning it with the reference teleontology (Schema.org), and the Data Layer, which ensures the consistency of values extracted from the source datasets.

4.1 Knowledge Layer

This subsection details the activities performed to formalize the vocabulary used in the Knowledge Graph. The process involved fixing the specific language terms for entities and attributes and subsequently aligning them with the reference Language Teleontology (Schema.org) to ensure semantic interoperability.

The initial terms derived from the raw datasets (often in Italian, e.g., Struttura Sanitaria, Farmacia) were standardised into English to create a consistent vocabulary. Concepts were also consolidated to reduce redundancy; for example, distinct datasets for Pharmacies and Parapharmacies were merged into the single term **Dispensary**, and Residential/Semi-Residential flows were unified under **CareStructure**.

The standardised terms were aligned with concepts from **Schema.org.** Where an exact match existed, the term was linked directly to the standard class. Where the standard ontology lacked sufficient granularity for the Trentino health domain, the Teleontology was enriched with new, purpose-specific classes.

1. Class: HealthFacility

- **Alignment:** This entity is aligned with `schema:MedicalOrganization` (`https://schema.org/MedicalOrganization`). It serves as the superclass for all healthcare providers in the graph.

- **Attribute Mapping:**
  - structureCode → `schema:identifier`
  - name → `schema:name`
  - address, CAP → `schema:address` (linking to `schema:PostalAddress`)
  - latitude, longitude → `schema:GeoCoordinates`
  - phone → `schema:telephone`
  - website → `schema:url`
  - email → `schema:email`
  - serviceName → `schema:name` (Contextual service description)

2. Class: Hospital

- **Alignment:** This entity is aligned with `schema:Hospital` (`https://schema.org/Hospital`). It inherits all attributes from `schema:MedicalOrganization`.
- **Attribute Mapping:**
  - ASL_Code → **New Data Property** (Created to store the Local Health Authority code).
  - managementType → **New Data Property** (Created to define the administrative management model).

3. Class: Dispensary

- **Alignment:** This entity is aligned with `schema:Pharmacy` (`https://schema.org/Pharmacy`). It represents both pharmacies and para-pharmacies.
- **Attribute Mapping:**
  - dispensaryType → `schema:additionalType` (Used to distinguish between 'Pharmacy' and 'Parapharmacy').
  - IVA (VAT Number) → `schema:vatID` (The fiscal identifier).

4. Class: CareStructure

- **Alignment: New Class**. This entity was created from scratch to model the specific "Residential and Semi-Residential" care facilities (RSA) found in the Trentino datasets. It is defined as a subclass of `schema:MedicalOrganization`.
- **Attribute Mapping:**
  - careType → `schema:additionalType` (Distinguishes 'Residential' vs 'Semi-Residential').
  - managementType → **New Data Property**.
  - **Statistical Attributes:** The following properties were created as new data properties to capture specific capacity metrics not available in Schema.org: `numBeds`, `numUsers`, `numAssistanceDays`, `numAdmissions`, `numDischarges`.

5. **Class: RehabilitationStructure**

- **Alignment: New Class**. Created to model therapeutic communities. It is defined as a subclass of `schema:MedicalOrganization`.
- **Attribute Mapping:**
  - ASL_Code → **New Data Property**.

6. **Class: Municipality**

- **Alignment:** This entity is aligned with `schema:AdministrativeArea` (`https://schema.org/AdministrativeArea`). It represents the territorial context.
- **Attribute Mapping:**
  - name → `schema:name`.
  - province → `schema:State`

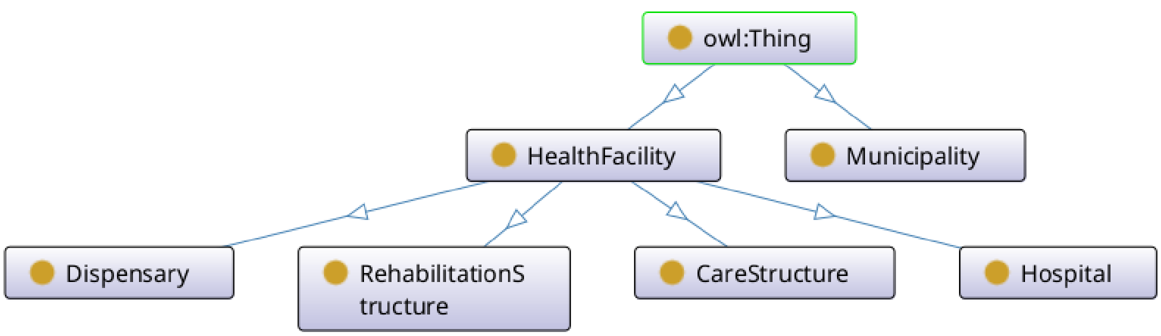

Figure 2: Ontology Language Definition

4.2 Data Layer

This section outlines the steps taken to clean, filter, and format the raw CSV datasets (`ASSRESIDENZIALE001`, `ASSSEMIRESIDENZIALE001`, `FARM001`, `OSPEDALI001`, `PARAFARM001`, `RIASTRUT001`, `SANSTRUT001`) into a usable format for the project.

4.2.1 Care Structures

Source Files: `ASSRESIDENZIALE001`, `ASSSEMIRESIDENZIALE001`
Script Used: `merging_careStructures.py`
We merged these two files into a single output file called **care structure**.

- **Assistance Days:** We merged different column names representing the “Number of Assistance Days” into one standard attribute.
- **Admissions/Discharges:** The first file included admission and discharge data, but the second did not. This resulted in some null values for these columns in the final file, which we treated as acceptable.
- **Cleanup:** We kept only the relevant attributes and removed (truncated) the rest.

4.2.2 Dispensaries

Source Files: `FARM001`, `PARAFARM001`
Script Used: `merging_dispensaries.py`
We combined pharmacies and parapharmacies into a single **dispensary** CSV.

- **Type Identification:** We added a new attribute called `DISPENSARY_TYPE` to distinguish between a pharmacy and a parapharmacy.
- **Contact Info:** Because our “Health Facility” structure requires attributes like phone, website, email, and `service_name`, but the source files didn’t have them, these fields were explicitly set to `NULL` to maintain structure.

4.2.3 Hospitals

Source Files: `OSPEDALI001`, `SANSTRUT001`
Script Used: `merging_hospitals.py`
We created a **HOSPITALS_MERGED** CSV by prioritizing the dedicated hospital file and supplementing it with data from the general health structure file.

- **Merging Logic:** We took all entries from `OSPEDALI001` and added any hospital from `SANSTRUT001` that was not a duplicate.
- **Services Attribute:** The `OSPEDALI001` file did not have an `ASSISTENZA` (Services) attribute, so those entries have null values. For entries coming from `SANSTRUT001`, we kept the services as a single joined string.

4.2.4 General Health Facilities

Source File: `SANSTRUT001`
Script Used: `cleaning_healthFacilities.py`
After extracting hospitals for the previous step, we cleaned the remaining data in `SANSTRUT001` to create the **Health Facility** CSV.

- **Filtering:** We removed all entries that were already moved to the Hospitals or Care Structures files.
- **Content:** This file now contains clinics, laboratories, and other non-hospital facilities. We kept only the main attributes and truncated the rest.

4.2.5 Manual Adjustments and Standardization

- **RIASTRUT001 Cleaning:** For this file, we manually removed two columns containing string values for longitude and latitude, as there was only one entry for them.
- **Language Normalization:** All attributes were renamed from Italian to English to match the project's requirements. This was done by hand to ensure the translations were accurate.
- **Municipality (COMUNE):** Instead of keeping the municipality data in a separate file, we added a `COMUNE` column directly to the CSV files to make the data easier to access.

4.3 Report

4.3.1 Entity Types (Etypes)

Initially, we identified 8 potential entity concepts from the raw datasets (e.g., distinguishing between Pharmacy and Parapharmacy, or Residential and Semi-Residential as separate types). After analysis in phase (1), these were consolidated into 6 final Entity Types to reduce redundancy and improve semantic clarity.

- Initially Considered: 8
- Finally Considered: 6
- Aligned to Teleontology: 4 (Directly), 2 (New/Custom)

Table 3: Entity Type (Etype) Term Definition and Alignment

Term	Init?	Final?	Alignment
HealthFacility	Yes	Yes	`schema:MedicalOrganization`
Hospital	Yes	Yes	`schema:Hospital`
Dispensary	Yes	Yes	`schema:Pharmacy`
CareStructure	Yes	Yes	-
Rehabilitation	Yes	Yes	-
Municipality	Yes	Yes	`schema:AdministrativeArea`
Service	Yes	No	-
Management	Yes	No	-

4.3.2 Object Properties

- **Initially Considered:** 4 terms
- **Finally Considered:** 2 terms
- **Aligned to Language Teleontology:** 2 terms

Refactoring Process The reduction in object properties was driven by a decision to flatten the ontology hierarchy. Services and Management Types were converted into **data properties**, removing their respective object properties. The remaining properties were renamed to match the target convention.

- **Data Property Integration:** The relations `hasService` and `is-managed-by` were removed; their data was integrated directly into Health Facilities, Hospitals, and Care Structures as attributes.
- **Renaming and Cardinality:** The property `locatedIn` was renamed to `has_Municipality`. The property `subclassOf` was renamed to `has_subclass` and specifically structured to link one parent Entity Type to four distinct child Entity Types.

Object Property Term Details The following table details the transformation of each object property term.

Table 4: Detailed alignment status of Object Property Terms.

Initial Term	Context (Domain/Range)	Action	Final Term
`hasService`	Health Facilities $\leftrightarrow$ Service	Removed	—
`is-managed-by`	Health Facilities $\leftrightarrow$ ManagementType	Removed	—
`locatedIn`	Health Facilities $\leftrightarrow$ Municipality	Renamed	`has_Municipality`
`subclassOf`	One EType $\rightarrow$ 4 ETypes	Renamed	`has_subclass`

4.3.3 Data Properties

- **Initially Considered:** 20 terms
- **Finally Considered:** 22 terms
- **Aligned to Language Teleontology:** 14 terms

Table 5: Detailed alignment status of Data Property Terms.

Attribute	Alignment
Entity: HealthFacility (Mapped to schema:MedicalOrganization)	
`structureCode`	`schema:identifier`
	Continued on next page...

Attribute	Alignment
name	schema:name
address	schema:streetAddress
CAP	schema:postalCode
latitude	schema:latitude
longitude	schema:longitude
phone	schema:telephone
website	schema:url
email	schema:email
serviceName	schema:name
Entity: Hospital (Mapped to schema:Hospital)	
ASLCode	—
ManagementType	—
Entity: RehabilitationStructure (New Class)	
ASLCode	—
Entity: Dispensary (Mapped to schema:Pharmacy)	
dispensaryType	schema:additionalType
IVA Type	schema:vatID
Entity: CareStructure (New Class)	
numBeds	—
numUsers	—
numAssistanceDays	—
numAdmissions	—
numDischarges	—
careType	schema:additionalType
ManagementType	—
Entity: Municipality (Mapped to schema:AdministrativeArea)	

Continued on next page...

Attribute	Alignment
`name`	`schema:name`
`province`	`schema:State`

5 Knowledge Definition

The Knowledge Definition phase represents the transition from the informal linguistic structures of our domain to a formal, machine-readable Knowledge Teleontology. Our primary objective was to refine the conceptual framework by enforcing semantic constraints, defining explicit domains and ranges for properties, and ensuring logical coherence across the model. By moving from simple terms to formalized ETypes with specific IRIs, we created a structure capable of supporting automated reasoning and data interoperability.

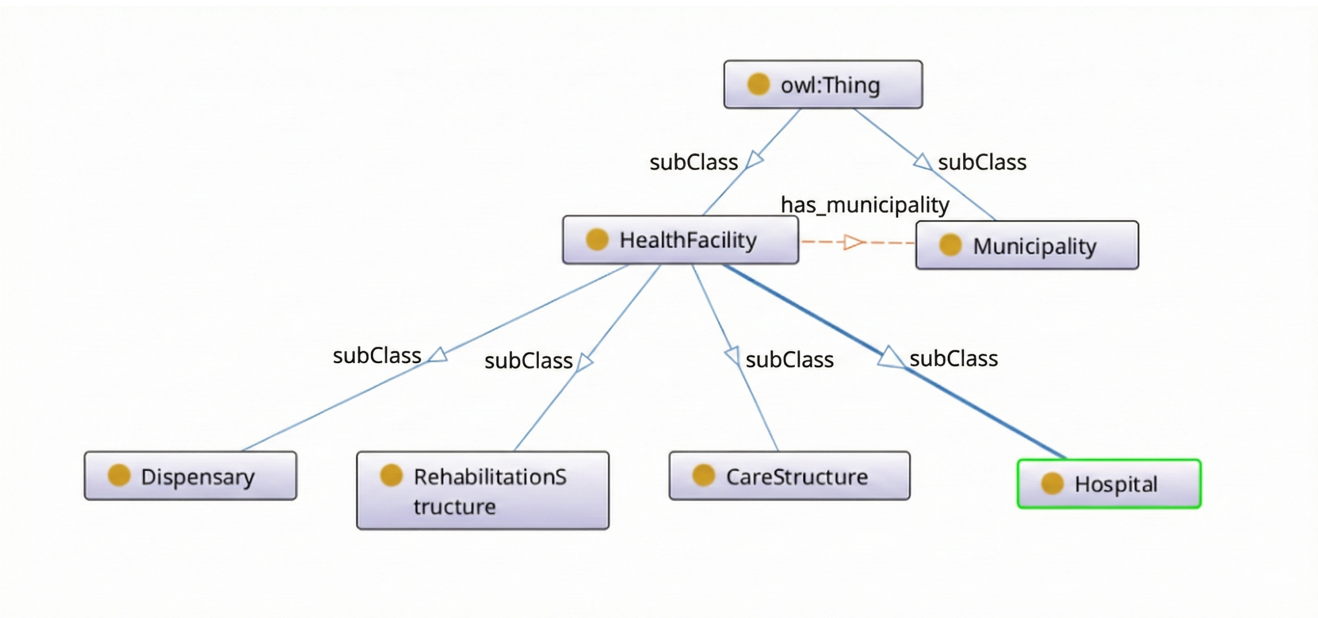

Figure 3: Formalized Knowledge Teleontology

5.1 Formalization of ETypes

The formalization began by establishing a root hierarchy under `owl:Thing`. Each entity was carefully categorized to balance global standards with local requirements.

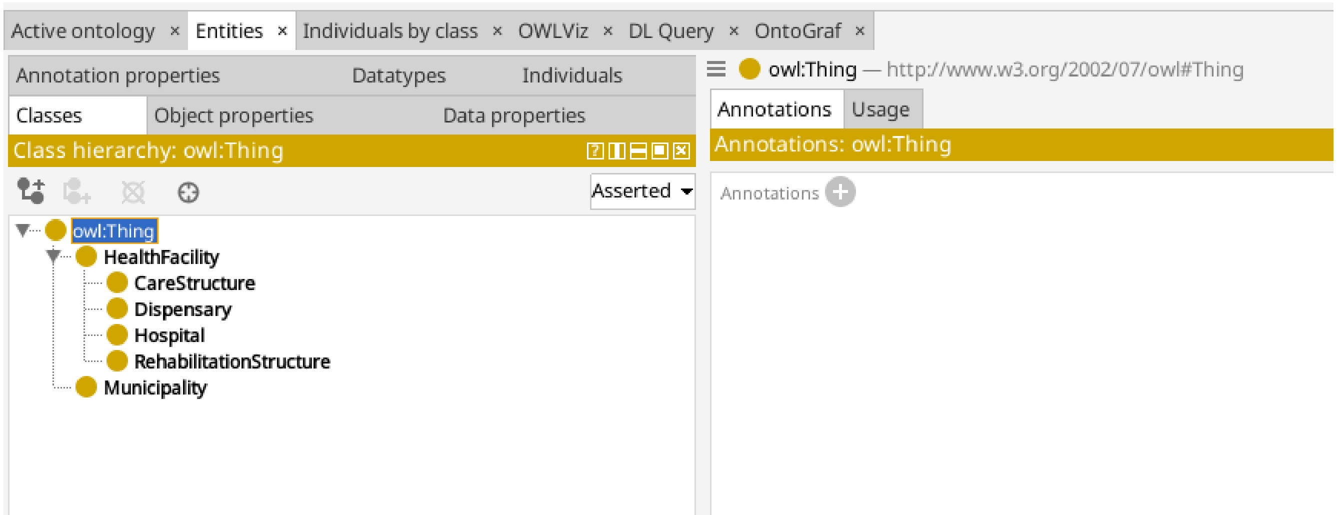

Figure 4: Hierarchical Structure of the ontology

- **Hierarchical Structure**: Entities like Hospital and CareStructure were defined as specialized subclasses of the HealthFacility superclass.
- **Namespace Standardization**: Every EType was assigned a unique IRI following the project convention: http://knowdive.disi.unitn.it/etype#<Name>.
- **Semantic Mapping**: We prioritized external alignment with the Schema.org vocabulary. For instance, HealthFacility was mapped to schema:MedicalOrganization to ensure the graph remains understandable to external systems.

5.2 Object Properties Formalization

Object properties were developed to serve as the relational glue of the ontology, formalizing the directional relationships between ETypes. These properties define how entities interact within the system:

- We established properties to link entities together, such as has_municipality, which creates a spatial link between a facility and its administrative territory.
- For each property, we specified the Domain and Range to indicate precisely which class the relationship originates from and which class it points to.
- Where possible, these properties were aligned with standard schemas; for example, our location relationship was mapped to schema:containedInPlace.

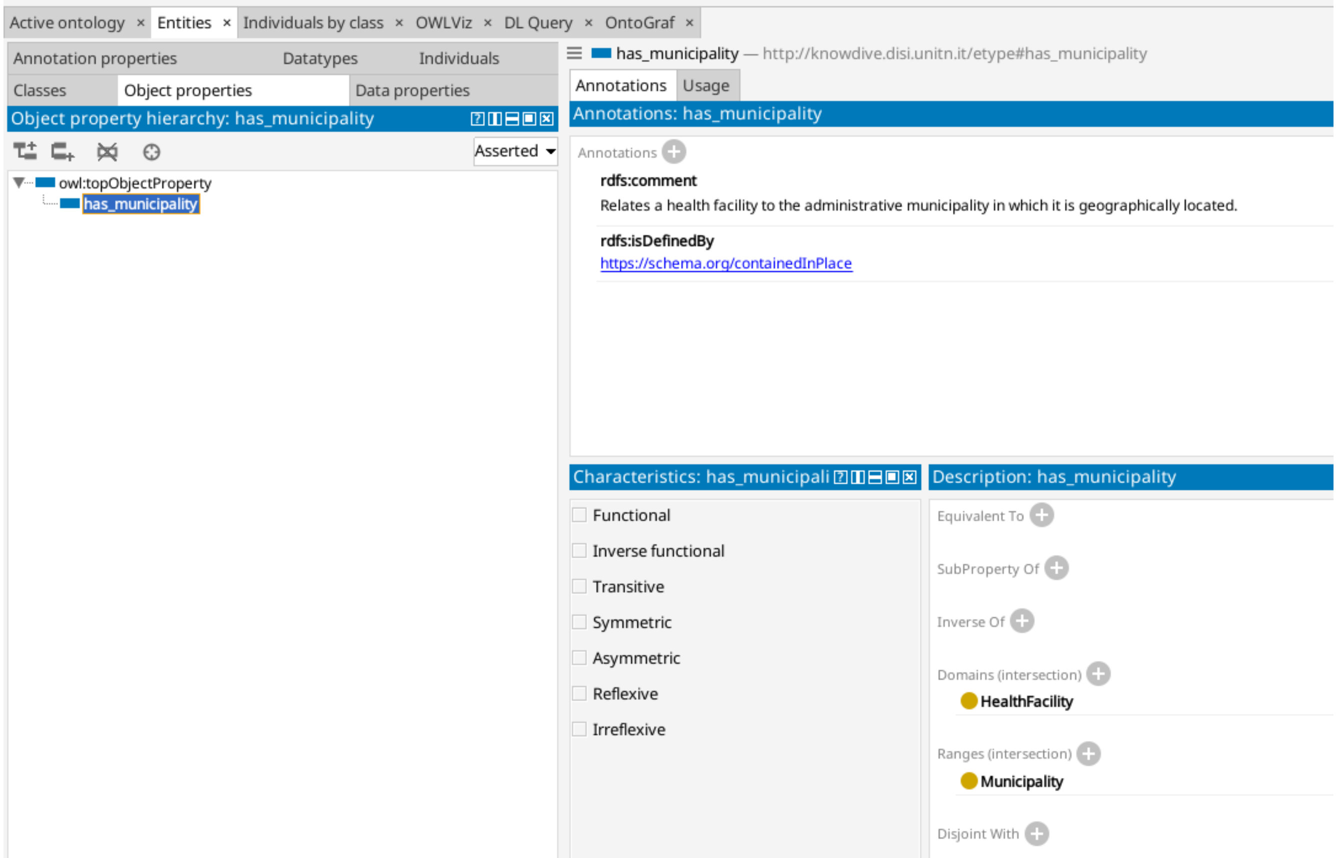

Figure 5: Object Properties Formalization

5.3 Data Properties Formalization

Data properties were created to capture the literal attributes and descriptive characteristics of each EType. This step ensures that every attribute is precisely constrained for validation:

- Following the ITELOS methodology, we standardized all data property names with the `has_` prefix to distinguish them from class names clearly.
- We assigned specific XML Schema datatypes (e.g., `xsd:string`, `xsd:integer`, `xsd:float`) to reflect the nature of the attribute, such as using floats for geographic coordinates and integers for bed counts.
- Each property was assigned to the specific EType that owns the attribute, ensuring that data like `has_numBeds` is correctly associated with medical structures.

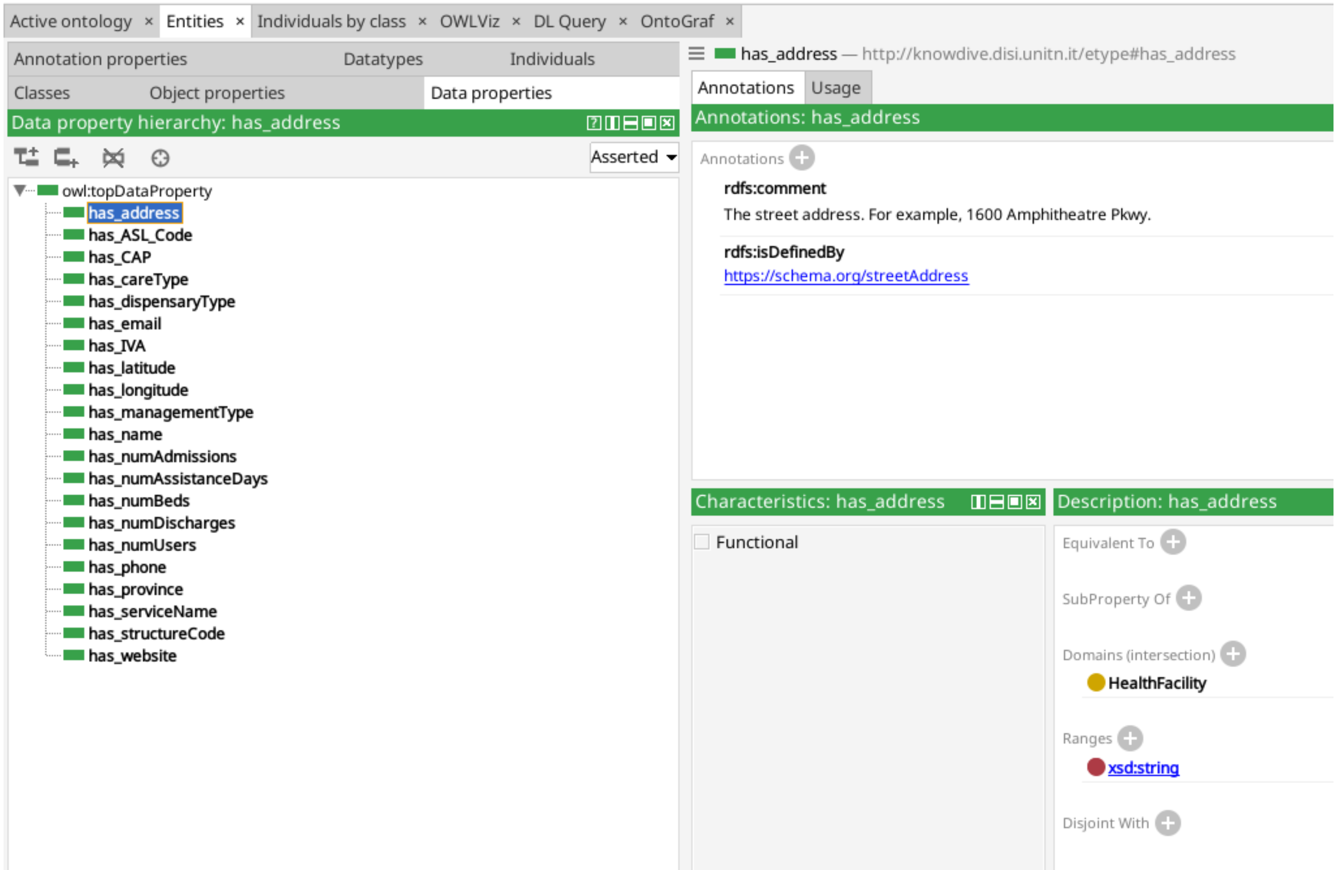

Figure 6: Data Properties Formalization

5.4 Report

This subsection addresses specific evaluative questions regarding the outcomes and evolution of the Knowledge Definition phase.

5.4.1 EType Evaluation

We initially considered 6 ETypes based on our language-level analysis. All 6 were retained in the final Knowledge Teleontology to ensure comprehensive coverage of the healthcare domain.

5.4.2 Object Properties

We initially identified 1 core object property to model geographical containment. This property was finalized as `has_municipality` and aligned with the `schema:containedInPlace` parent property.

5.4.3 Data Properties

We initially considered 21 data properties to capture administrative and quantitative details. All 21 were retained with no major change.

5.4.4 Data Cleaning

We noticed that some entries from hospital.csv dataset were not removed from HealthFacilities.csv dataset due to them having a slight change in the name of suffix like "S.P.A". these were removed manually due to the low number of them(6 entries).

6 Entity Definition

The **Entity Definition** phase represents the operational core of the iTelos methodology, where the previously defined Knowledge Layer (teleontology) and Data Layer (cleaned datasets) are merged into a unified, machine-interpretable Knowledge Graph (KG). During this phase, the distinction between knowledge and data activities dissolves as we populate the ontological structures with concrete values from our source files to form a single data structure.

6.1 Entity Identification and Data Mapping

The first set of activities focused on individual datasets, transforming flat CSV records into a relational RDF structure using the Karma Data Integration tool. This involved two simultaneous sub-activities:

- **Entity Identification**: We assigned unique, persistent Internationalized Resource Identifiers (IRIs) to each entity instance. We selected the `structureCode` as the primary identifier. By designating this column as the `uri of Class` and ensuring the radio button was correctly selected, we prevented the creation of transient "Blank Nodes" (*genids*) observed in early mapping iterations.
- **Data Mapping**: We linked data values to their corresponding ontological properties. For instance, facility names were mapped to the `has_name` data property, while coordinates were linked to `has_latitude` and `has_longitude`.
- **Data Enrichment**: Since provincial data was missing from source files, we used a Python Transform (`return "Trentino"`) in Karma to inject a virtual column. This was mapped as an attribute of the `Municipality` entity rather than the facility, maintaining the correct administrative hierarchy.

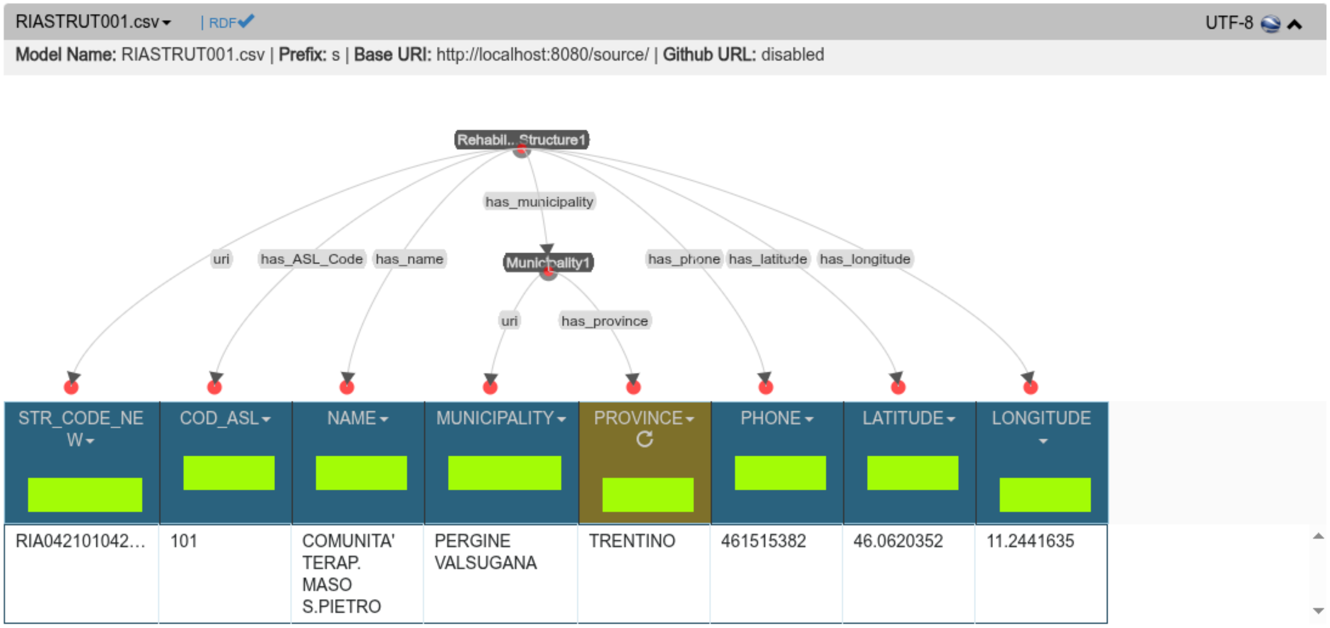

Figure 7: Entity Definition in Karma.

6.2 Phase Outcomes and Validation

The final outcome is a structured Knowledge Graph in Turtle (.ttl) format. We validated the result using the GraphDB Visual Graph tool, confirming that the relationships—such as the nested provincial attributes—were correctly serialized and that all entities were dereferenceable via their assigned IRIs rather than anonymous nodes.

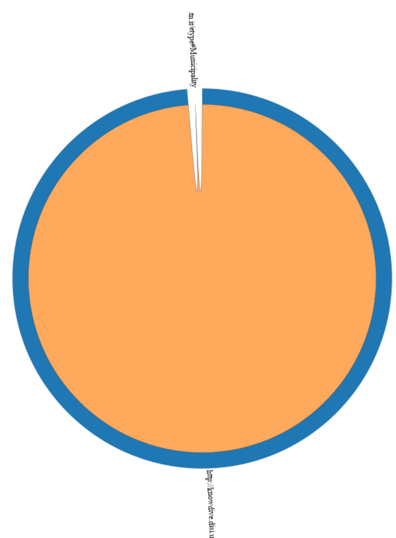

Figure 8: Class Relationships in GraphDB

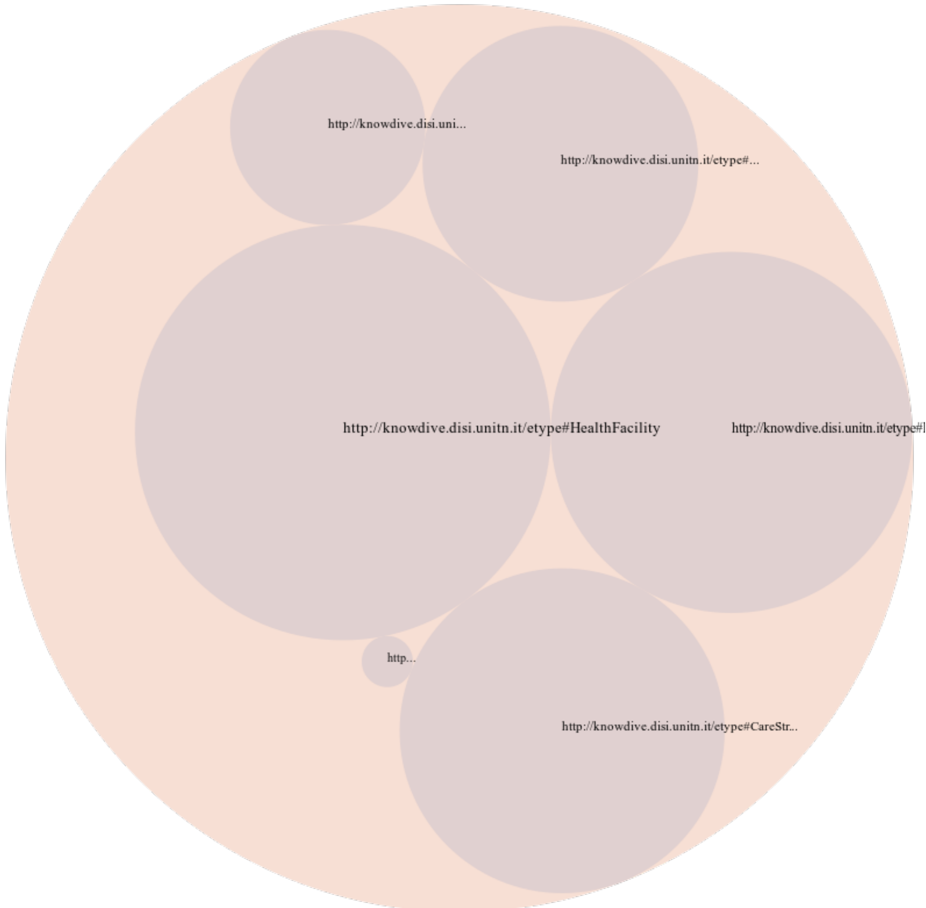

Figure 9: Class Hierarchy in GraphDB

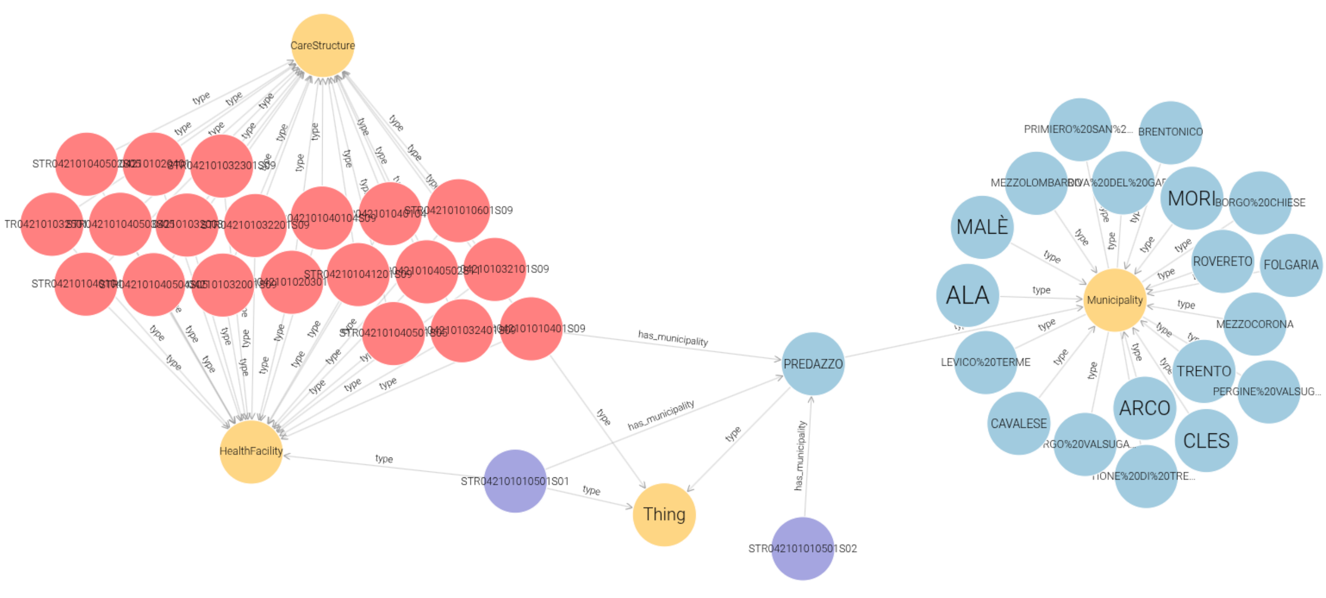

Figure 10: Visual Graph in GraphDB

6.3 Validation through Competency Question Execution

A critical outcome of the Entity Definition phase is the functional validation of the Knowledge Graph via SPARQL query execution. We translated the natural language Competency Questions into formal queries to verify that the integrated data layer correctly populates the knowledge layer defined in preceding phases.

The following queries demonstrate the retrieval of information across the relational structure of the graph, specifically testing the object property links between facilities and municipalities.

6.3.1 CQ 1.2: Find address and phone number for a specific facility.

Listing 1: SPARQL query for CQ 1.2

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
SELECT ?name ?address ?phone
WHERE {
    ?facility a etype:HealthFacility ;
              etype:has_name ?name ;
              etype:has_address ?address ;
              etype:has_phone ?phone .
    FILTER(CONTAINS(?name, "OSPEDALE DI TRENTO"))
}
```

name	address	phone
"OSPEDALE DI TRENTO"	"LARGO MEDAGLIE D ORO, 9"	"461903015"

Table 6: Results for CQ 1.2

6.3.2 CQ 2.1 Residential facilities for Elderly Assistance in a municipality.

Listing 2: SPARQL query for CQ 2.1

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>
SELECT ?facilityName
WHERE {
    ?facility a etype:HealthFacility ;
              etype:has_name ?facilityName ;
              etype:has_careType "Residential" ;
              etype:has_serviceName ?service ;
              etype:has_municipality ?municipalityURI .

    FILTER(CONTAINS(?service, "ANZIANI"))
    FILTER(STR(?municipalityURI) = "http://localhost:8080/source/PREDAZZO")
}
```

facilityName
"AZIENDA PUBBLICA SERVIZI ALLA PERSONA - RSA PREDAZZO"

Table 7: Results for CQ 2.1

6.3.3 CQ 4.3 Health Structures Managed Directly

Listing 3: SPARQL query for CQ 4.3

```
PREFIX etype: <http://knowdive.disi.unitn.it/etype#>

SELECT (COUNT(?facility) AS ?managedCount)
WHERE {
    ?facility etype:has_municipality <http://localhost:8080/source/TRENTO> ;
              etype:has_managementType ?type .

    FILTER(CONTAINS(?type, "DIRETTAMENTE␣GESTITA"))
}
```

managedCount
"30"

Table 8: Results for CQ 4.3

6.4 Report

The Entity Definition phase successfully integrated the conceptual and data layers without altering the initial entity count. A key refinement included the manual enrichment of source datasets with a `Province` column to support territorial modeling within the `Municipality` entity.

Technical difficulties regarding URI serialization and tool interface stability were resolved through iterative mapping cycles in Karma, ensuring that all entities were correctly identified by stable IRIs rather than anonymous nodes. The successful validation in GraphDB confirms that the Knowledge Graph is structurally sound and consistent with all preceding methodology phases.

During a small inspection to the ontology, we noticed an addition data property "has_hospitalType" which was not available in the dataset nor was utilized in any sense, so we manually deleted it. This change led to a remodeling in the ER diagram and minor adjustments to the Purpose Definition and Language Definition phases.